DAY 38 · THEORY HANDOUT

Build an Agent That Uses Tools

Let an agent manage the model-tool loop, with a clear stopping limit.

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The idea and its useful terms

Let an agent manage the model-tool loop, with a clear stopping limit.

Agent

A model-driven loop that can choose tools.

State

The messages and values kept during the run.

Stop condition

A rule that ends work or limits repeated steps.

A fixed chain follows an order we choose. An agent can choose a tool and next step from the current messages.

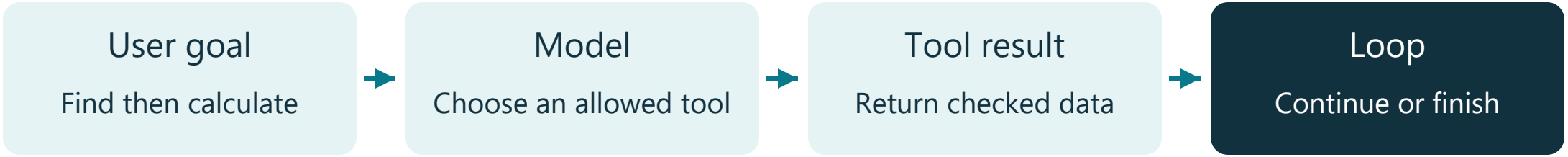
create_agent builds the loop around the model. Tools still run as real Python functions and need clear inputs and limits.

> An agent is not a guarantee of good judgment. A tool-capable model and tested tools are needed.

Follow the work, one step at a time

Each step has a job. We can inspect its output before moving on.

Read the path from left to right.



Each arrow means: pass the result to the next step.

create_agent connects model decisions with tool execution and messages. The loop continues after a tool result until the model finishes or a limit stops the run. Read the messages to see which tools it actually used.

A fixed chain follows our chosen path. An agent chooses its next step from available tools. Model-selection middleware can choose a different model, but message count alone does not prove a question is harder.

> If the answer is wrong, inspect the input and the intermediate results. The last sentence alone cannot show which step failed.

A small example we can check

The values here are classroom examples. The notebook shows actual results.

Goal

Two notebooks cost 40 each.

What is the total?

Tool

multiply 2 by 40.

Returns 80.

Final reply

Total: 80.

Check the tool trace.

Read the given input first. Point to the fact or value needed for the requested output.

Run the matching example, then try the fresh input at the end of the notebook. Compare both results with the known input.

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Choose the setting or tool for its job

A different choice changes the work. These are the choices we compare.

values stream

Show the full state after each step.

updates stream

Show only changed parts. Useful to follow tool work.

Model selection

Use a tested model. Message count is not quality.

Build one agent with add and multiply tools.

Inspect the actual messages and tool results, not just the last sentence.

> Run a fresh two-step question and set a recursion limit to stop repeated loops.

Practice, check, and read the full lesson

Use the notebook to try the steps. Use the article when you need the longer explanation.

1. Build

Run the small example.

Inspect each step.

2. Change

Try the fresh input.

Keep the goal clear.

3. Check

Compare with the source.

Keep mistakes visible.

Run a fresh two-step question and set a recursion limit to stop repeated loops.

Read the complete tutorial breakdown for this section in our KGP Talkie article.

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DAY 38 · COMPLETE

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